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ORIGINAL ARTICLE

Comparing PSA Screening Patterns and Their Role as Predictor of Prostate Cancer Diagnosis: Analysis of a Contemporary North American Cohort

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Received: 23 November 2024 | **Revised:** 6 January 2025 | **Accepted:** 13 January 2025

Funding: This study is supported by the Vattikuti Foundation.

Keywords: middle aged | mortality | prostate-specific antigen | prostatic neoplasms

ABSTRACT

Introduction: PSA screening remains a pivotal tool for early prostate cancer (PCa) detection. International guidelines rely on evidence from three major randomized clinical trials: ERSPC, PLCO, and CAP. We aim to examine the percentage of patients in real-world practice who get PSA screening as defined by each of the aforementioned trials. Moreover, we seek to evaluate if the different PSA screening patterns have a different impact on PCa incidence and its features at diagnosis.

Materials and Methods: Our institutional database was queried to identify men aged 55–69 who received at least one PSA test, did not develop PCa or die within 6 years of the initial test, had follow-up within our system at least 6 years after the initial test, and did not have a previous PCa diagnosis. A total of 28,612 patients met our selection criteria. We categorized patients into three distinct PSA screening patterns based on testing frequency (PLCO: 1 PSA test per year for 6 years; ERSPC: 2 or 3 PSA tests over 6 years; CAP: 1 PSA test over 6 years). Our primary outcomes were any PCa incidence and clinically significant PCa (csPCa, defined as ISUP ≥ 3) incidence. Secondary outcome was the rate of cM1 disease. Competing risks cumulative incidence curves were used to depict any PCa and csPCa diagnosis with death before a diagnosis considered a competing risk. Multivariable competing risks regression (CRR) was used to assess the impact of the different screening patterns on any PCa and csPCa incidence, after adjusting for confounding factors.

Results: The most prevalent PSA screening pattern was ERSPC, including 15,530 patients (54.3%), followed by the CAP with 9003 patients (31.5%), and the PLCO with only 4079 patients (14.2%). The median (IQR) follow-up time was 4.8 (1.7–10.8) years. At 10 years, any PCa incidence was 7.4% versus 5.6% versus 2.5% for PLCO versus ERSPC versus CAP, respectively, while for csPCa, the rates were 2.5% versus 2.5% versus 1.2% (both $p < 0.001$). On multivariable analyses, PLCO and ERSPC patterns were associated with 2.92-fold and 2.31-fold higher risks from 1 year to the next of any PCa diagnosis, respectively, compared to CAP pattern (both $p < 0.001$). Similarly, patients with PLCO and ERSPC patterns had 2.07-fold and 2.31-fold higher risks, respectively, of csPCa diagnosis compared to CAP pattern (both $p < 0.001$). In men with PCa diagnosis, the rates of cM1 disease were respectively 1.7% vs 5.6% vs 10.8% for PLCO versus ERSPC versus CAP, respectively ($p = 0.0009$).

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Conclusion: We observed that the most common screening pattern in "real-world" clinical practice is close to what ERSPC recommend, and this pattern seems to achieve a reasonable reduction in the risk of advanced PCa, while limiting overdiagnosis.

1 | Introduction

Prostate cancer (PCa) is the most common non-cutaneous cancer and the second leading cause of cancer death in men within the United States [1]. Serum prostate-specific antigen (PSA) screening aims to identify PCa at an early, treatable stage and has been shown to reduce prostate cancer-specific mortality (PCSM) [2, 3]. As such, PSA screening remains a pivotal tool for early PCa detection, despite also having been linked to a significant increase in over-diagnosis and over-treatment [4, 5].

International guidelines regarding PSA screening are heavily relying on evidence derived from three major randomized clinical trials: the European Randomized Study of Screening for Prostate Cancer (ERSPC), the Prostate, Lung, Colorectal, and Ovarian Cancer Screening Trial (PLCO), and the Cluster Randomized Trial of PSA Testing for Prostate Cancer (CAP) [6–8]. However, these trials have focused on different age range and utilized different PSA screening frequencies. Thus, despite its widespread use, the optimal frequency of PSA screening remains a subject of debate, and it is also unclear how often patients actually undergo PSA testing in the real-world scenario.

To fill this gap, we aimed to examine the percentage of patients in real-world practice who get PSA screening, as defined by each of the aforementioned trials. Moreover, we sought to evaluate if the different PSA screening patterns (as defined by the aforementioned trials) had a different impact on PCa diagnosis and the stage of the disease at diagnosis.

2 | Materials and Methods

2.1 | Data Source

We utilized our institutional database which was built by trained programmers who interrogating our electronic medical records for all men receiving care in Henry Ford Health System between 1995 and 2022. Of note, our system of over 50 medical centers and hospitals serves the Detroit metropolitan population which is vastly diverse in its social and racial components. We focused on men aged between 55 years and 69 years who received at least one PSA test, fit into one of the three screening patterns of interest, did not develop PCa or die within 6 years of the initial test, and who had follow-up data available at least 6 years after the initial PSA test. These selection criteria resulted in a total of 28,612 patients (Figure 1).

2.2 | Variables

Our main variable of interest was PSA screening pattern. We categorized patients into three distinct PSA screening patterns based on testing frequency over 6 years past their initial PSA test. In doing so, we outlined three groups, namely PLCO pattern (1 PSA test per year for 6 years), ERSPC pattern (2 or 3 PSA tests over 6 years) and CAP pattern (1 PSA test over 6 years). We also extracted the following variables: race, first PSA value (at study entry), age at first PSA, year of first PSA, Charlson Comorbidity Index (CCI) at study entry [0, 1, ≥ 2], and ADI (Area Deprivation Index) National Quartile. For patients with PCa diagnosis, we also extracted age at diagnosis, clinical TNM stage, highest Gleason Grade, initial gland treatment [radical prostatectomy, radiation therapy, other] and Hormonal treatment status [yes vs. no].

2.3 | Endpoints

We had two main primary endpoints: (1) any PCa diagnosis and (2) clinically significant PCa (csPCa) diagnosis, defined as International Society of Urological Pathology (ISUP) grade ≥ 3. Our secondary endpoint was the rate of cM1 at diagnosis.

2.4 | Statistical Analysis

Descriptive statistics consisted of medians with interquartile ranges (IQR) for continuous variables while frequencies and percentages were reported for categorical variables. Patients were then stratified based on PSA screening pattern (PLCO vs. ERSPC vs. CAP), and the three groups were compared by ANOVA for continuous covariates and chi-square tests for categorical covariates. Competing risks cumulative incidence curves were then used to depict the rate of any PCa and that of csPCa in the entire cohort, with death before a PCa diagnosis considered a competing risk, after stratifying men based on the screening pattern they received. Multivariable competing risks regression (CRR) was used to investigate the association between PSA screening patterns and the risk of developing any PCa and csPCa diagnosis, after adjusting for all available covariates. The number of events for the secondary endpoint was limited, so we reported only descriptive analysis for that endpoint as no further analyses were feasible. Two-sided p values < 0.05 indicated statistical significance. All analyses were performed using SAS 9.4 (SAS Institute, Cary, North Carolina). Patients were retrospectively enrolled in HFH-IRB#12641-29. An institutional review board waiver for informed consent was obtained prior to conducting this study, in accordance with institutional regulations when dealing with de-identified previously collected data.

3 | Results

Descriptive statistics of the cohort are presented in Table 1. Overall, the most prevalent PSA screening pattern observed was the ERSPC, including 15,530 patients (54.3%), followed by CAP with 9003 patients (31.5%), and PLCO with only 4079 patients (14.2%). Most patients were White (71.6%) and had a CCI of 0 (39.1%). Within our cohort, the median age at the first PSA test was

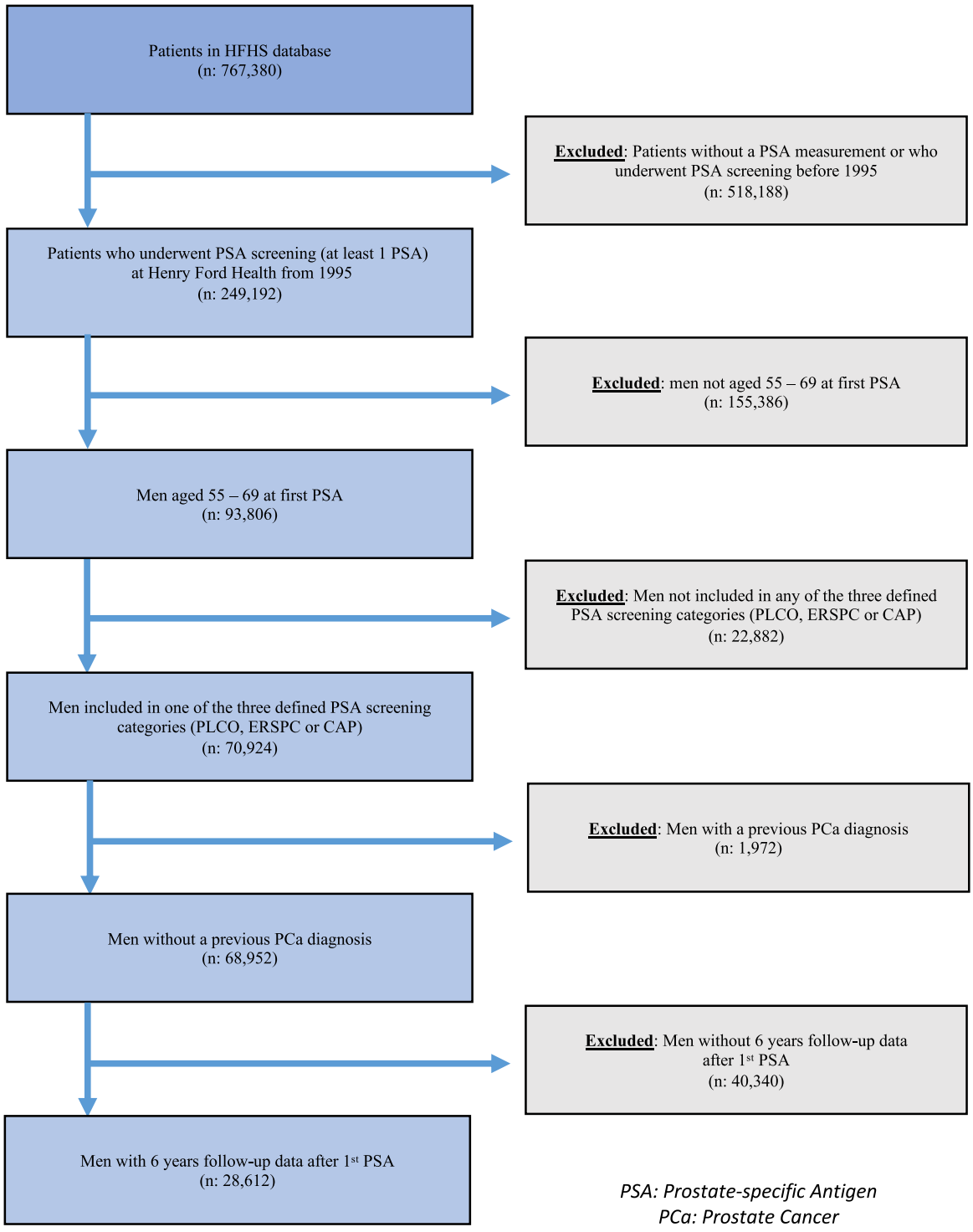


FIGURE 1 | Inclusion/exclusion criteria resulting in 28,612 men aged between 55 years and 69 years who received at least one PSA test and did not develop PCa or died within 6 years of the initial test, within the Henry Ford Health System. [Color figure can be viewed at wileyonlinelibrary.com]

61 (IQR: 57–65), and the median first PSA was 1 (IQR: 0.5–1.8) ng/mL. Comparing the three populations, the percentage of Non-Hispanic Black patients was higher (23.1%) in the less intensive screening pattern (CAP) group. Similarly, the percentage of patients included within the highest (and most socio-demographically disadvantaged) ADI quartile was higher (34.6%) in the less intensive screening pattern (CAP) group. Additionally, the percentage of patients with CCI of 0 was higher (43.9%) in the most intensive screening pattern (PLCO) group (all $p < 0.001$). The median (IQR) follow-up time was 4.8 (1.7–10.8) years. Within this follow-up, 1108 (3.9%) patients were diagnosed with PCa and 483 (1.7%) patients were diagnosed with csPCa. Among patients diagnosed with PCa, 62 (5.6%) had cM1 at diagnosis.

On competing risks cumulative incidence analysis, for the PLCO, ERSPC and CAP patterns, the 10-year any PCa diagnosis rates were respectively 7.4%, 5.6%, and 2.5% ($p < 0.001$) (Figure 2). For the same groups, the 10-year csPCa incidence

TABLE 1 | Descriptive statistics of 28,612 patients who received at least 1 PSA test from the ages of 55–69 did not die or develop PCa within 6 years of the initial PSA test and had follow-up at least 6 years after the initial PSA test.

	Screening pattern				
	PLCO (1 PSA test per year for 6 years) (*N* = 4079)	**ERSPC (2 or 3 PSA tests over 6 years) (*N* = 15530)**	**CAP (1 PSA test over 6 years) (*N* = 9003)**	**Total (*N* = 28,612)**	***p* value**
First PSA (at study entry)					< 0.0001[a]
N (Missing)	4079 (0)	15,530 (0)	9003 (0)	28,612 (0)	
Median (IQR)	1.2 (0.6, 2.7)	1.0 (0.5, 1.7)	0.9 (0.5, 1.8)	1.0 (0.5, 1.8)	
Age at first PSA					< 0.0001[a]
N (Missing)	4079 (0)	15,530 (0)	9003 (0)	28,612 (0)	
Median (IQR)	63.0 (59.0, 66.0)	61.0 (57.0, 65.0)	60.0 (57.0, 65.0)	61.0 (57.0, 65.0)	
Year of first PSA					< 0.0001[a]
N (Missing)	4079 (0)	15,530 (0)	9003 (0)	28,612 (0)	
Median (IQR)	2005 (1996, 2014)	2008 (1998, 2014)	2008 (2000, 2014)	2008 (1998, 2014)	
Age at diagnosis					0.0116[a]
N (Missing)	239 (3840)	693 (14,837)	176 (8827)	1108 (27,504)	
Median (IQR)	73.0 (69, 76)	72 (68, 76)	71 (67, 75)	72 (68, 76)	
Race, *n* (%)					< 0.0001[b]
Black	663 (16.3%)	2890 (18.6%)	2080 (23.1%)	5633 (19.7%)	
White	3157 (77.4%)	11,327 (72.9%)	6010 (66.7%)	20494 (71.6%)	
Other	121 (3.0%)	590 (3.8%)	366 (4.1%)	1077 (3.8%)	
Unknown	138 (3.3%)	723 (4.7%)	547 (6.1%)	1408 (4.9%)	
CCI (at study entry), *n* (%)					< 0.0001[b]
0	1792 (43.9%)	6173 (39.7%)	3215 (35.7%)	11,180 (39.1%)	
1	476 (11.7%)	2226 (14.3%)	1120 (12.4%)	3822 (13.4%)	
2	265 (6.5%)	804 (5.2%)	430 (4.8%)	1499 (5.2%)	
3+	91 (2.2%)	454 (2.9%)	323 (3.6%)	868 (3.0%)	
Unknown	1455 (35.7%)	5873 (37.8%)	3915 (43.5%)	11,243 (39.3%)	
ADI National Quartile, *n* (%)					< 0.0001[b]
1%–24%	520 (12.7%)	1972 (12.7%)	1217 (13.5%)	3709 (13.0%)	
25%–49%	1141 (28.0%)	3642 (23.5%)	1897 (21.1%)	6680 (23.3%)	
50%–74%	1310 (32.1%)	4748 (30.6%)	2446 (27.2%)	8504 (29.7%)	
75%+	1003 (24.6%)	4673 (30.1%)	3118 (34.6%)	8794 (30.7%)	
Unknown	105 (2.6%)	495 (3.2%)	325 (3.6%)	925 (3.2%)	
Clinical T Stage, *n* (%)					0.1315[b]
cT1	171 (71.5%)	467 (67.4%)	123 (69.9%)	761 (68.7%)	
cT2	33 (13.8%)	121 (17.5%)	27 (15.3%)	181 (16.3%)	
cT3	0 (0.0%)	10 (1.4%)	5 (2.8%)	15 (1.4%)	
cT4	0 (0.0%)	6 (0.9%)	0 (0.0%)	6 (0.5%)	
cTx	35 (14.6%)	89 (12.8%)	21 (11.9%)	145 (13.1%)	
No PCa diagnosis	3840	14837	8827	27,504	

(Continues)

TABLE 1 | (Continued)

	Screening pattern				
	PLCO (1 PSA test per year for 6 years) (*N* = 4079)	**ERSPC (2 or 3 PSA tests over 6 years) (*N* = 15530)**	**CAP (1 PSA test over 6 years) (*N* = 9003)**	**Total (*N* = 28,612)**	***p* value**
Clinical N Stage, *n* (%)					0.0029[b]
cN0	126 (52.7%)	334 (48.2%)	76 (43.2%)	536 (48.4%)	
cN1	1 (0.4%)	25 (3.6%)	13 (7.4%)	39 (3.5%)	
cNx	112 (46.9%)	334 (48.2%)	87 (49.4%)	533 (48.1%)	
No PCa diagnosis	3840	14,837	8827	27,504	
Clinical M Stage, *n* (%)					0.0009[b]
cM0	165 (69.0%)	471 (68.0%)	122 (69.3%)	758 (68.4%)	
cM1	4 (1.7%)	39 (5.6%)	19 (10.8%)	62 (5.6%)	
cMx	70 (29.3%)	183 (26.4%)	35 (19.9%)	288 (26.0%)	
No PCa diagnosis	3840	14,837	8827	27504	
Highest Gleason Grade, *n* (%)					0.0133[b]
< =6	87 (36.4%)	179 (25.8%)	37 (21.0%)	303 (27.3%)	
3 + 4	70 (29.3%)	194 (28.0%)	58 (33.0%)	322 (29.1%)	
4 + 3	35 (14.6%)	121 (17.5%)	29 (16.5%)	185 (16.7%)	
8	15 (6.3%)	71 (10.2%)	12 (6.8%)	98 (8.8%)	
9/10	22 (9.2%)	79 (11.4%)	27 (15.3%)	128 (11.6%)	
Unknown	10 (4.2%)	49 (7.1%)	13 (7.4%)	72 (6.5%)	
No PCa diagnosis	3840	14,837	8827	27,504	
PCa status, *n* (%)					< 0.0001[b]
No PCa diagnosis	2705 (66.3%)	10,367 (66.8%)	5658 (62.8%)	18,730 (65.5%)	
PCa diagnosis	239 (5.9%)	693 (4.5%)	176 (2.0%)	1108 (3.9%)	
Death before PCa Dx	1135 (27.8%)	4470 (28.8%)	3169 (35.2%)	8774 (30.7%)	
CS PCa status, *n* (%)					< 0.0001[b]
No CS PCa Diagnosis	2862 (70.2%)	10,740 (69.2%)	5753 (63.9%)	19,355 (67.6%)	
CS PCa Diagnosis	82 (2.0%)	320 (2.1%)	81 (0.9%)	483 (1.7%)	
Death Before PCa Dx	1135 (27.8%)	4470 (28.8%)	3169 (35.2%)	8774 (30.7%)	
Time to PCa diagnosis or follow-up (years)					< 0.0001[a]
N (Missing)	4079 (0)	15,530 (0)	9003 (0)	28,612 (0)	
Median (IQR)	5.7 (1.9, 13.4)	4.6 (1.7, 10.9)	4.8 (1.8, 9.7)	4.8 (1.7, 10.8)	
Initial treatment (in patients diagnosed with PCa), *n* (%)					< 0.0001[b]
Radical prostatectomy	74 (31.0%)	219 (31.6%)	64 (36.4%)	357 (32.3%)	
Radiation therapy	92 (38.4%)	187 (27.0%)	44 (25%)	323 (29.1%)	
Other	73 (30.6%)	287 (41.4%)	68 (38.6%)	428 (38.6%)	
Hormonal treatment status (in patients diagnosed with PCa), *n* (%)					0.1617[b]
No hormone therapy	191 (79.9%)	557 (80.4%)	152 (86.4%)	900 (81.2%)	
Hormone therapy	48 (20.1%)	136 (19.6%)	24 (13.6%)	208 (18.8%)	

[a]Kruskal–Wallis *p* value.
[b]Chi-square *p* value.

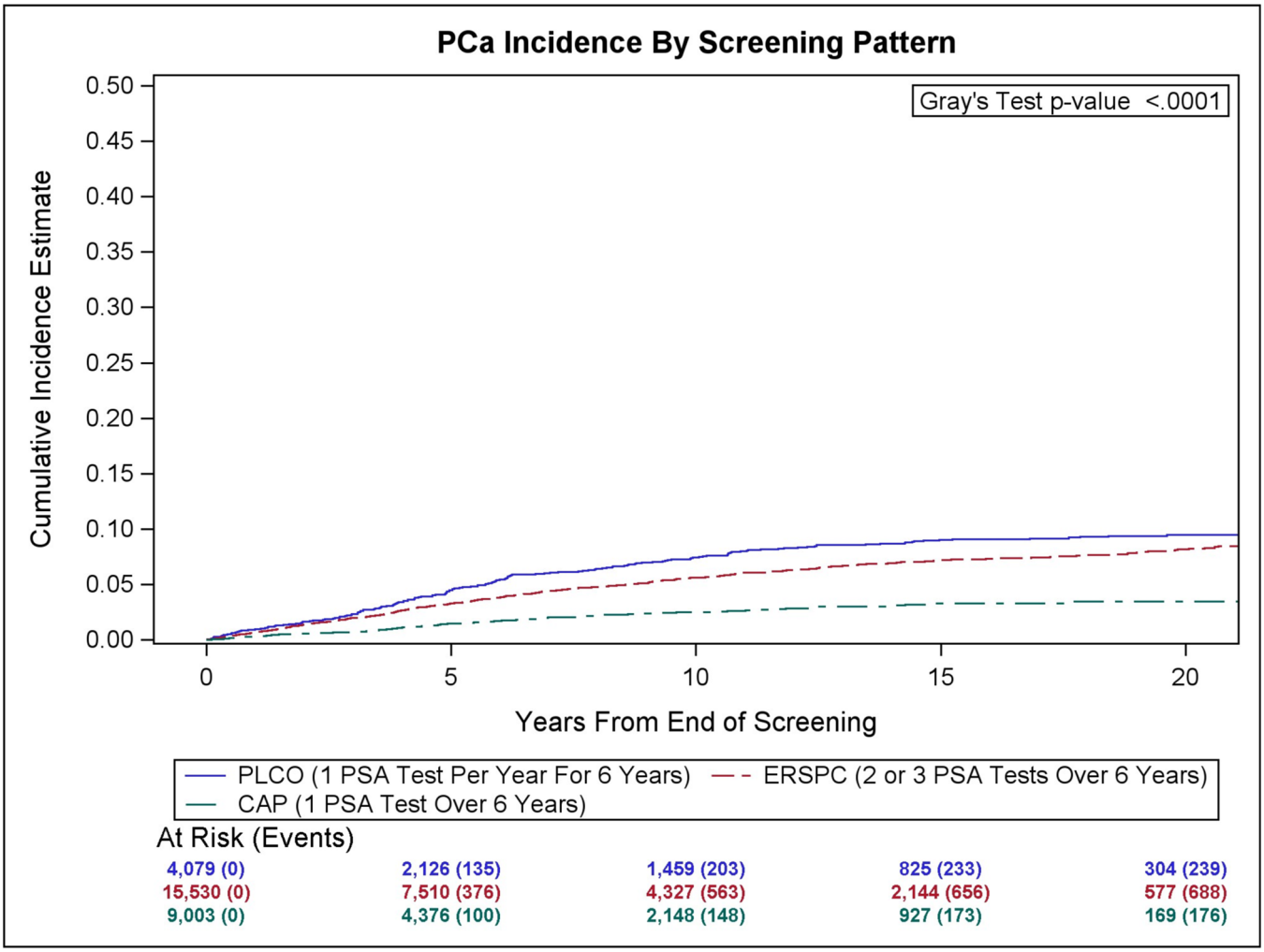


FIGURE 2 | Cumulative incidence curve for any prostate cancer diagnosis by screening pattern in 28,612 men who received at least one PSA test and did not develop PCa or died within 6 years of the initial test, within the study period from 1995 to 2022. [Color figure can be viewed at wileyonlinelibrary.com]

rates were respectively 2.5%, 2.5%, and 1.2% ($p < 0.001$) (Figure 3). In men with PCa diagnosis, the rates of cM1 at diagnosis were respectively 1.7%, 5.6%, and 10.8% ($p = 0.0009$) (Table 1).

On multivariable CRR analysis, patients who underwent PSA screening in accordance with the PLCO and ERSPC patterns had a 2.92- and 2.31-fold higher risks of receiving any PCa diagnosis, respectively, compared to patients who underwent PSA screening in accordance to the CAP pattern (both $p < 0.001$) (Table 2). Likewise, patients who underwent PSA screening in accordance to the PLCO and ERSPC patterns had a 2.07- and 2.31-fold higher risks of receiving csPCa diagnosis, respectively, compared to patients who underwent PSA screening in accordance to the CAP pattern (both $p < 0.001$) (Table 3).

4 | Discussion

Different PSA screening trials used different PSA screening frequencies and reported contradictory outcomes. To date, there is very limited literature about the frequency of PSA screening in "real-world" (non-trial) setting, and its relation to PCa diagnosis. In our study, we sought to understand the prevalence of different PSA screening patterns (as suggested by PLCO, ERSPC, and CAP) and their subsequent impact on the PCa diagnosis.

Several of our findings are worth highlighting. First, examining distinct patterns in the utilization of PSA screening, we observed that most patients (54.3%) were screened in a pattern comparable to ERSPC, while as many as 31.5% were screened opportunistically (as in the CAP trial). Interestingly, only 14.2% of our population underwent PSA screening according to the PLCO pattern (1 PSA test per year for 6 years). To the best of our knowledge, our report is the first to examine the prevalence of PLCO, ERSPC, and CAP screening pattern in a "real-world" North American cohort. Interestingly, we found that most men adhered to the European (ERSPC), and not the American (PLCO) screening pattern. That said, our analysis does not allow a further dive into the cause of this phenomenon, which could be the result of physician preferences, patients compliance, access to care variability or a combination of the aforementioned, as well as other causes.

Second, the analysis of cumulative incidence revealed interesting insights into the differential impact of PSA screening patterns. Specifically, we observed that more intensive screening

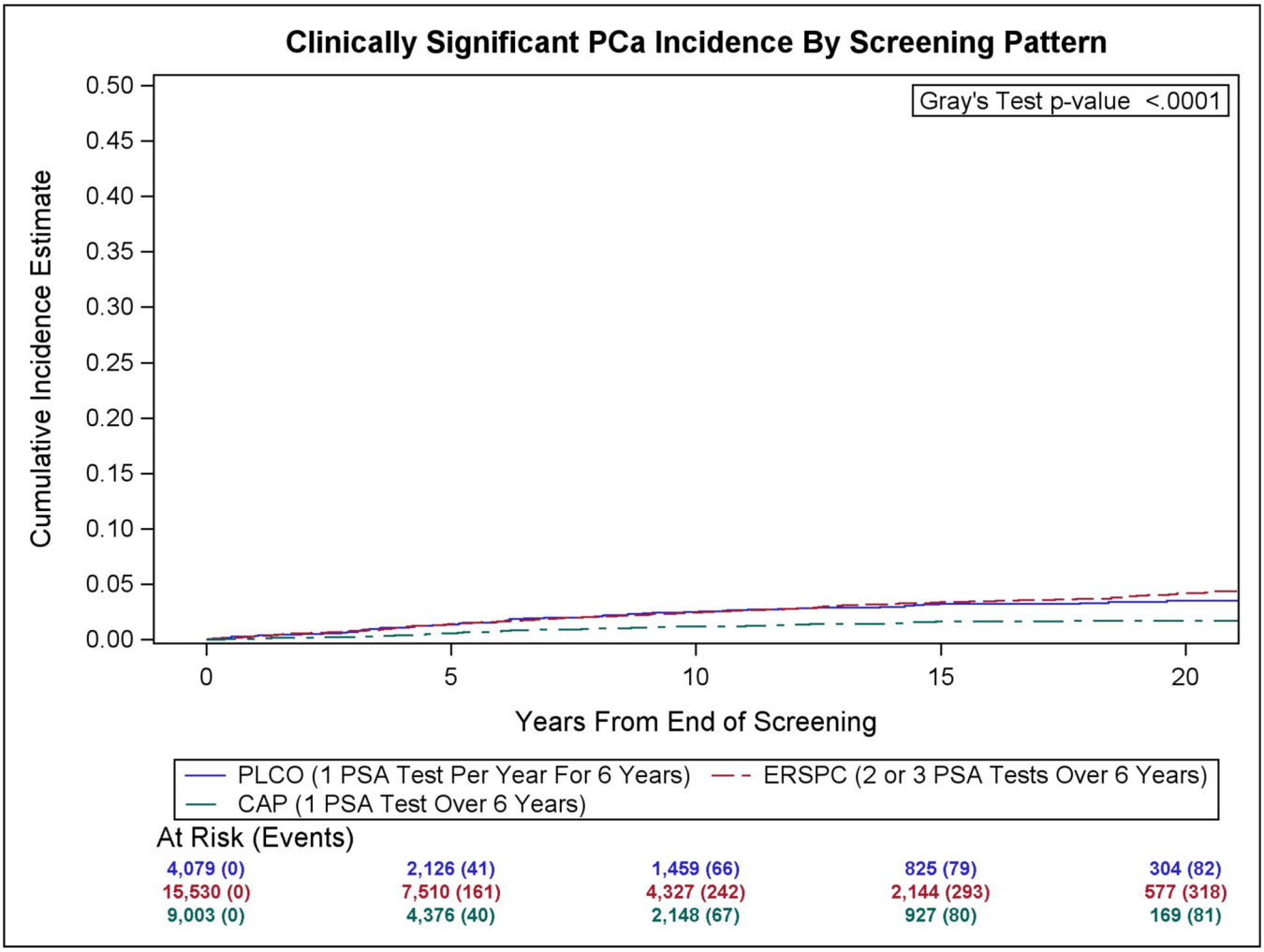


FIGURE 3 | Cumulative incidence curve for Clinically significant Prostate Cancer diagnosis by screening pattern in 28,612 men who received at least one PSA test and did not develop PCa or died within 6 years of the initial test, within the study period from 1995 to 2022. [Color figure can be viewed at wileyonlinelibrary.com]

patterns (PLCO and ERSPC) were associated with higher rates of PCa diagnosis. The PLCO pattern achieved the highest number of PCa diagnoses in terms of any PCa (almost 3 times compared with the CAP pattern and 1.5 times compared with ERSPC). In contrast, when focusing on csPCa, the PLCO and ERSPC patterns diagnosed twice as many cases of PCa as the CAP pattern. Our findings seem to corroborate the previous literature about the already known trade-off between intensive PSA testing and potential overdiagnosis: although it was conducted in an older population, a study by Shao et al. showed how an intensive PSA testing (defined as 4 to 6 PSA tests in 5 years) was likely to increase the risk of overdiagnosis and overtreatment among elderly men [9].

Third, in the group subjected to more intensive PSA screening (PLCO), we observed a significantly lower rate of cM1 disease at diagnosis (1.7%). This finding aligns with the notion that frequent and rigorous PSA screening contributes to the early detection of prostate cancer, facilitating intervention at localized stages. The reduced rate of metastatic disease in this cohort underscores the potential effectiveness of proactive screening measures in identifying prostate cancer at its earlier, more treatable stages. On the contrary, in the group with the least frequent PSA screening (CAP), we identified the highest rate of cM1 disease at diagnosis (10.8%). This finding underscores the potential consequence of limited or sporadic screening efforts, resulting in delayed detection and an increased likelihood of metastatic disease. The elevated incidence of advanced-stage PCa in this cohort emphasizes the critical role of consistent and timely screening practices in mitigating disease progression. Consistent with our findings, evidence from a longitudinal randomized controlled trial indicates that repeated PSA testing is correlated with a decreased risk of advanced-stage disease [10]. Specifically, The Finnish Randomized Study of Screening for Prostate Cancer revealed that individuals screened only once had the highest incidence of high-risk PCa. In contrast, those who underwent two or three PSA screenings experienced a reduced risk of advanced PCa [10]. Another recent study by Aktary et al. that sought to analyze the Health-Related and Psychosocial Factors Associated with PCa stage at diagnosis in a Canadian cohort showed that a higher number of PSA tests over a lifetime was significantly protective against advanced-stage diagnosis [11].

Of note, we lack specific data on the use of pre-biopsy multiparametric MRI (mpMRI) and the biopsy technique employed

TABLE 2 | Fine-Gray MVA predicting any prostate cancer diagnosis in 28,612 men who received at least one PSA test and did not develop PCa or died within 6 years of the initial test.

		Years from end of screening		
Covariate	**Level**	**Hazard ratio**	**HR *p* value**	**Type3 *p* value**
Screening pattern	ERSPC (2 or 3 PSA tests over 6 years)	2.31 (1.96–2.72)	**<0.001**	**<0.001**
	PLCO (1 PSA test per year for 6 years)	2.92 (2.40–3.56)	**<0.001**	
	CAP (1 PSA test over 6 years)	—	—	
Age at first PSA		0.95 (0.94–0.97)	**<0.001**	**<0.001**
Race	Black	1.86 (1.62–2.13)	**<0.001**	**<0.001**
	Other	1.10 (0.79–1.53)	0.558	
	Unknown	0.86 (0.63–1.17)	0.336	
	White	—	—	
CCI at first PSA	1	0.62 (0.51–0.76)	**<0.001**	**<0.001**
	2	0.45 (0.32–0.63)	**<0.001**	
	3+	0.53 (0.34–0.83)	**0.006**	
	Unknown	0.70 (0.61–0.80)	**<0.001**	
	0	—	—	
ADI national quartile	25%–49%	0.96 (0.79–1.16)	0.654	**0.045**
	50%–74%	0.96 (0.79–1.16)	0.664	
	75%+	0.83 (0.69–0.99)	**0.040**	
	Unknown	1.25 (0.90–1.73)	0.180	
	1%–24%	—	—	
First PSA (ng/mL)		1.00 (1.00–1.00)	0.540	0.540
Year of first PSA		0.98 (0.97–0.99)	**<0.001**	**<0.001**

Note: Bold values indicate statistically significant.

(systematic vs. targeted fusion biopsy). Recent studies have demonstrated that mpMRI and targeted biopsy techniques significantly improve diagnostic accuracy, reducing both missed diagnoses of clinically significant cancer and overdiagnosis of indolent disease [12]. The lack of consideration of these aspects might have influenced the observed rates of diagnosis and disease stage. However, our "real-world" cohort approach aims to reflect everyday clinical practices, which often include variability in diagnostic pathways.

Fourth, we observed that non-Hispanic Black patients and those in the most disadvantaged ADI quartile group received less frequent screening. Previous literature can explain these two findings that can be considered tightly linked to each other. For numerous neoplasms, including PCa, socioeconomic status has been shown to be inversely correlated with incidence and mortality rates [13]. Moreover, many earlier works ascribed racial differences in PCa to access to care issues rather than genetic differences in the nature of the disease [14, 15]. An even more detailed aspect of the matter was provided by a study from Kinlock et al., which described a higher percentage of non-Hispanic Black men reporting medical mistrust, compared to non-Hispanic White men [16].

Although determining which of the patterns described is the best in absolute terms is not the goal of this work, arguably, considering the data regarding PCa diagnosis and the cM1 rate at diagnosis, the ERSPC pattern could be considered as a fair compromise between the PLCO (more intense) pattern and the CAP (less intense) pattern. To the best of our knowledge, our study is the first to examine screening patterns in a real-world setting. It bridges the gap between randomized trials and clinical practice, also assessing the predictive value of distinct screening patterns on two important cancer control outcomes.

Our work's limitations include the retrospective nature of the study. Additionally, unknown confounders might also have affected our observations. A further potential limitation is that our cohort might fail to capture PSA testing performed outside of our health system; this limitation is partially mitigated by the approximately 50 locations within the health system that offer laboratory services. Finally, an important limitation of our study is the lack of specific data on the diagnostic pathway followed for each patient. We were unable to determine whether mpMRI was used prior to biopsy or which biopsy technique (systematic vs. targeted fusion biopsy) was employed. These factors can significantly influence prostate cancer diagnosis, particularly the detection of clinically significant cancer. Our study's strength includes a large and racially diverse population. Many earlier studies relied on self-reported PSA screening, which has limited accuracy and may vary by race, while we relied on laboratory PSA values within our health system's electronic medical record.

TABLE 3 | Fine-Gray MVA predicting clinically significant prostate cancer diagnosis in 28,612 men who received at least one PSA test and did not develop PCa or died within 6 years of the initial test.

		Time to CS PCa diagnosis or follow-up (years)		
Covariate	**Level**	**Hazard ratio**	**HR *p* value**	**Type3 *p* value**
Screening pattern	ERSPC (2 or 3 PSA tests over 6 years)	2.31 (1.81–2.94)	**< 0.001**	**< 0.001**
	PLCO (1 PSA test per year for 6 years)	2.07 (1.51–2.83)	**< 0.001**	
	CAP (1 PSA test over 6 years)	—	—	
Age at first PSA		0.97 (0.95–0.99)	**0.015**	**0.015**
Race	Black	1.92 (1.56–2.36)	**< 0.001**	**< 0.001**
	Other	0.98 (0.57–1.68)	0.941	
	Unknown	1.13 (0.75–1.71)	0.549	
	White	—	—	
CCI at first PSA	1	0.58 (0.43–0.80)	**< 0.001**	**< 0.001**
	2	0.49 (0.29–0.81)	**0.006**	
	3+	0.64 (0.35–1.19)	0.158	
	Unknown	0.75 (0.62–0.92)	**0.006**	
	0	—	—	
ADI National Quartile	25%–49%	1.05 (0.78–1.42)	0.757	0.914
	50%–74%	1.06 (0.80–1.43)	0.674	
	75%+	0.95 (0.72–1.26)	0.734	
	Unknown	0.95 (0.53–1.68)	0.849	
	1%–24%	—	—	
First PSA (ng/mL)		1.00 (1.00–1.00)	0.500	0.500
Year of first PSA		0.97 (0.95–0.99)	**0.001**	**0.001**

Note: Bold values indicate statistically significant.

5 | Conclusion

Our study is the first to examine screening patterns in a real-world setting and to assess the predictive value of distinct screening patterns on PCa incidence. We observed that the most common screening pattern in "real-world" clinical practice is close to what ERSPC recommend, and this pattern seems to achieve a reasonable reduction in the risk of advanced PCa, while limiting overdiagnosis.

Acknowledgments

The VCORE is supported by a fund started by a contribution from the Menon Foundation and the VUI Foundation.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

I hereby declare that data supporting the results of this study are available upon reasonable request to the corresponding author [F.A.].

The data are not publicly available due to restrictions because they contain information that could compromise the privacy of research participants.

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